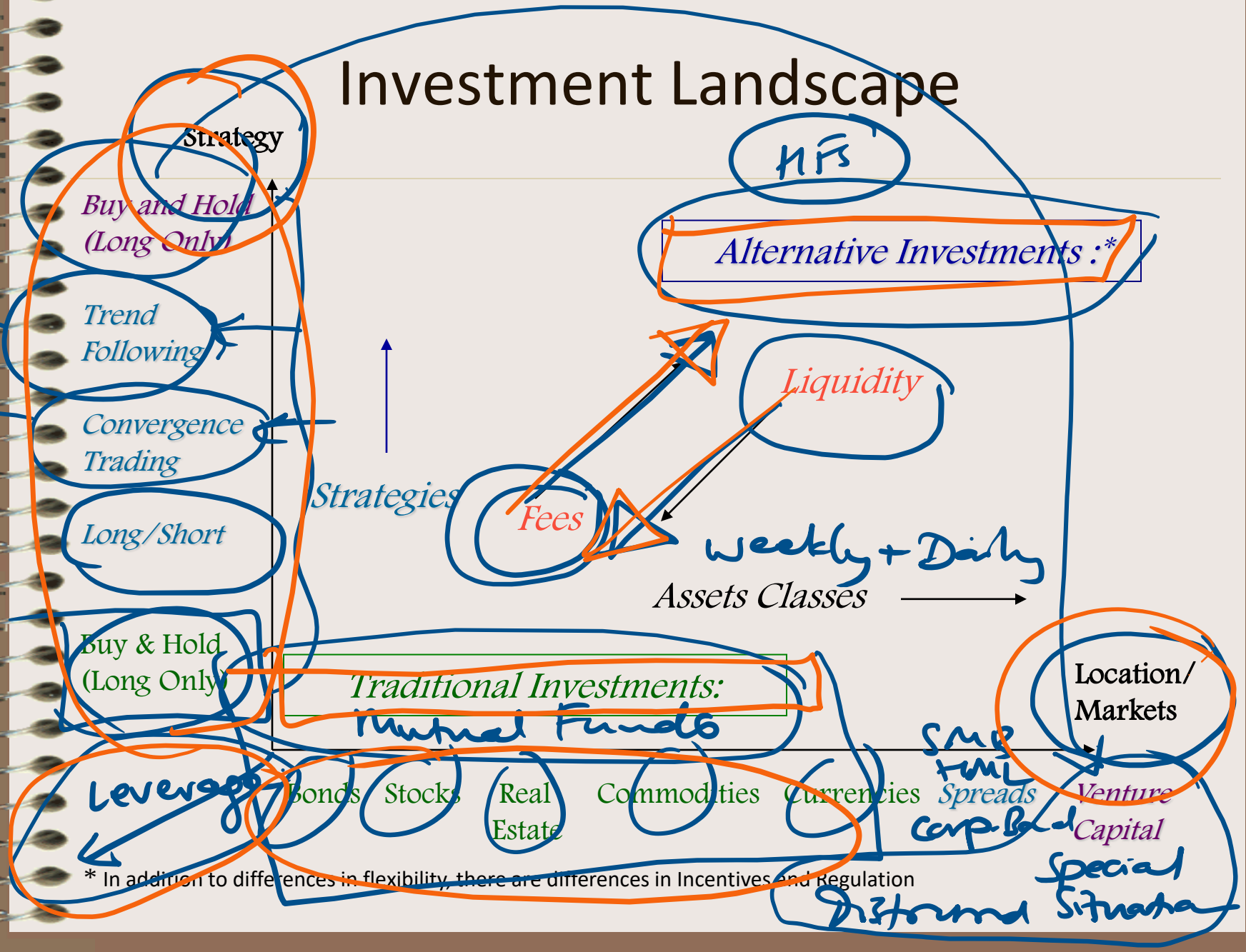
Hedge Funds: The Past, The Present and The Future?*

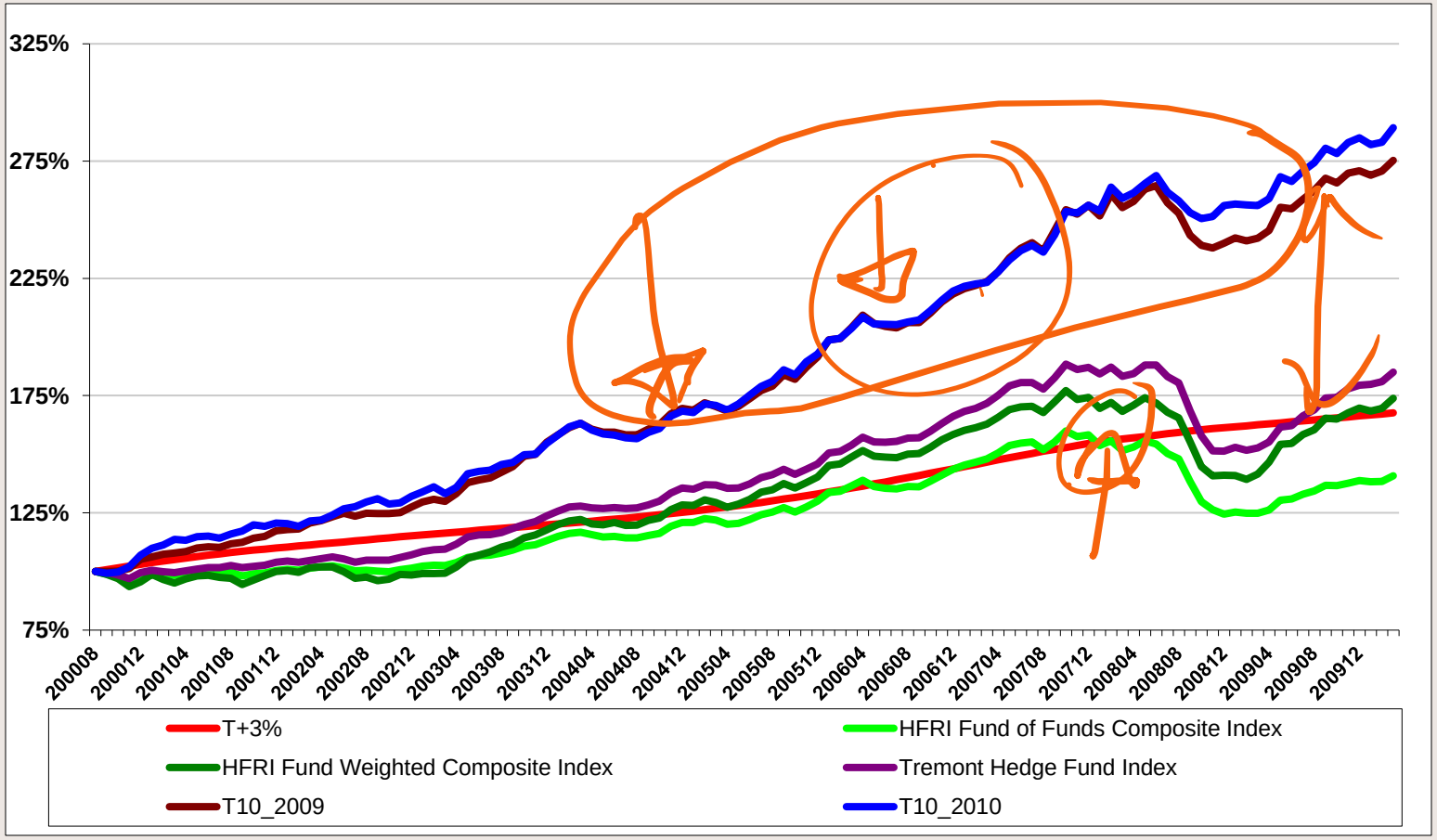
Prof. Narayan Y. Naik
London Business School

Investment Landscape



* In addition to differences in flexibility, there are differences in Incentives and Regulation

Performance of Hedge Fund Indices versus that of the Flagship Funds of the "Blue Chip" firms



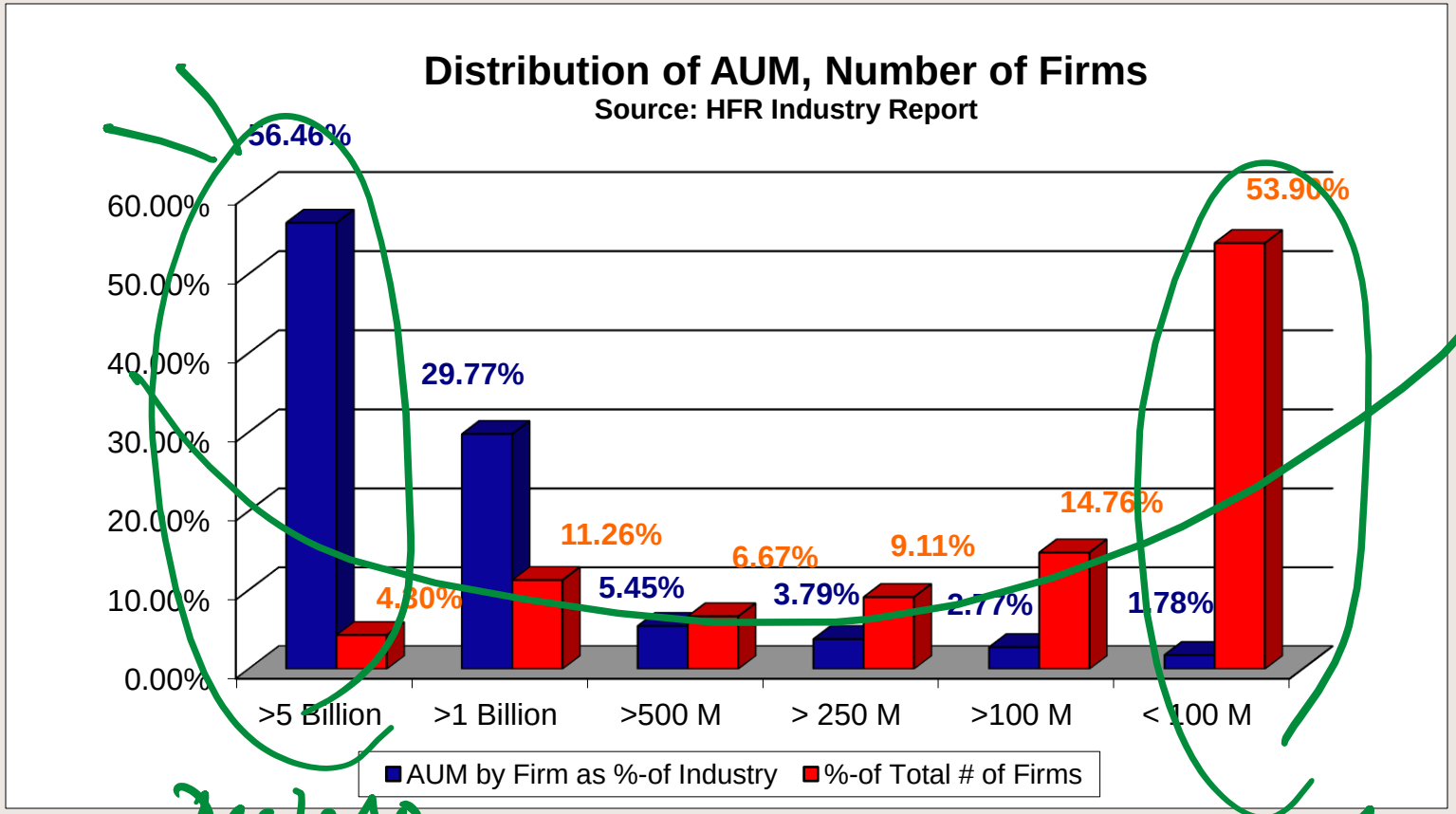
Capital Structure of the Industry
Distribution of AUM versus Number of Firms

Q? Do investors place more emphasis on the company/manager (driver) or the strategy/fund (car)

•In order t

histo

Capital Structure of the Industry
Distribution of AUM versus Number of Firms



Decile 10

Decile 1

Insights from Research:

Hedge Funds and Managed Futures can add value to a Balanced Portfolio

Returns to Manager Skill:

What investors are willing to pay 2/20 for (or 3/30 for FoHFs)

α

Returns to Systematic Risk Factors:

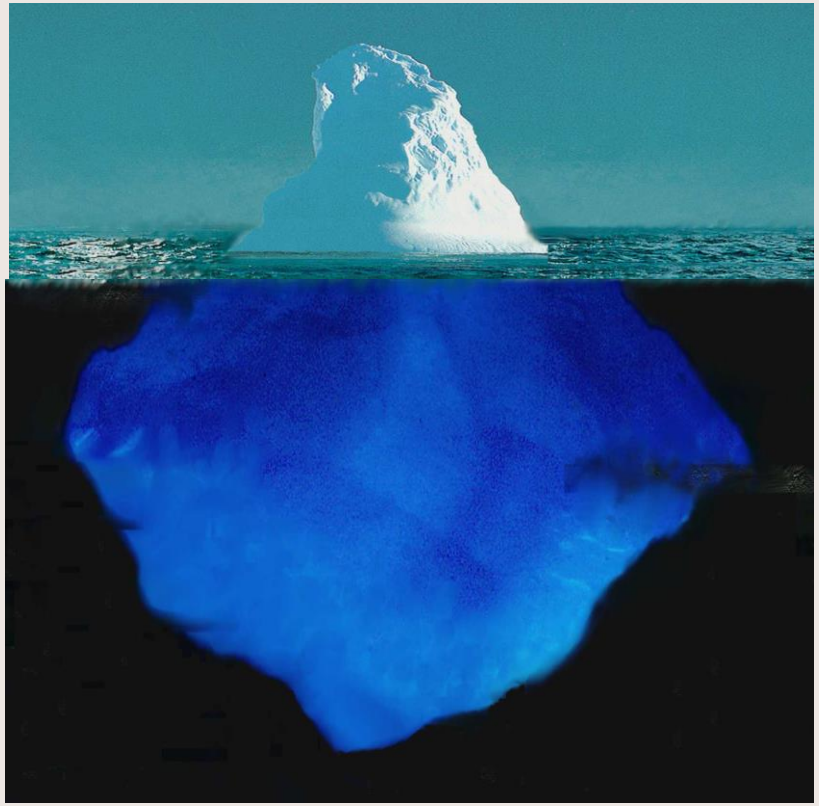
Exotic Betas

β

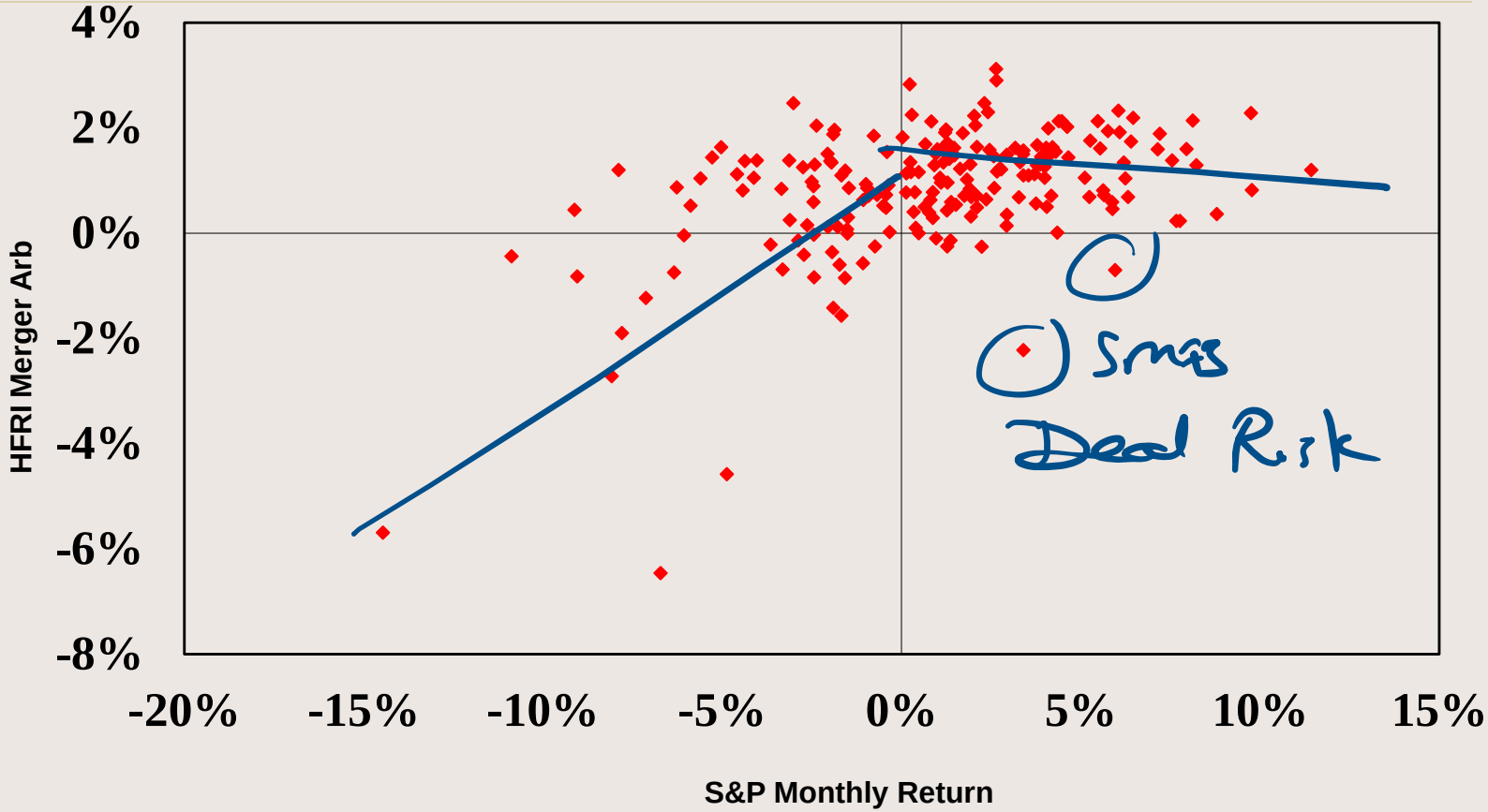
Alternative Betas

Traditional Betas

β

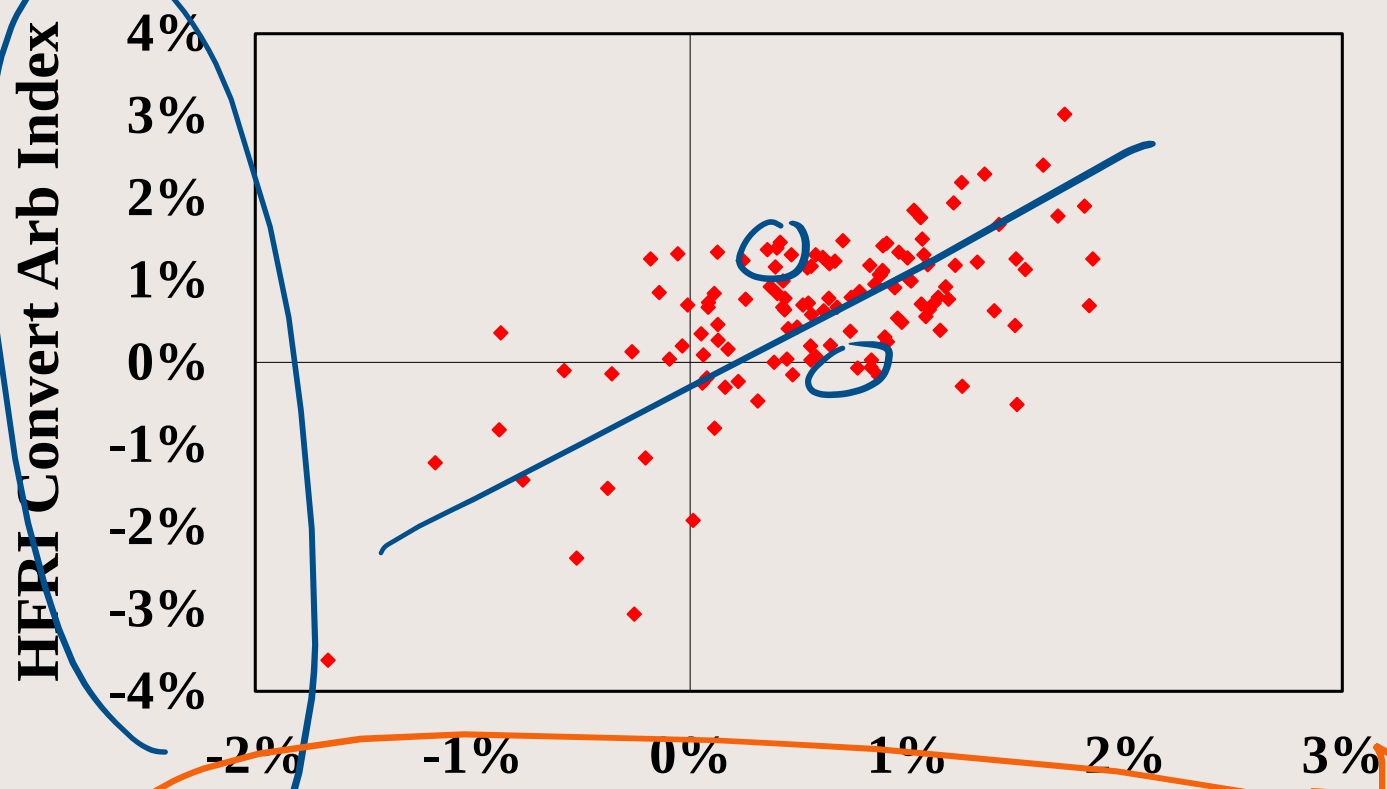


Merger Arbitrage



Source: Datastream, HFR (1990-2006)

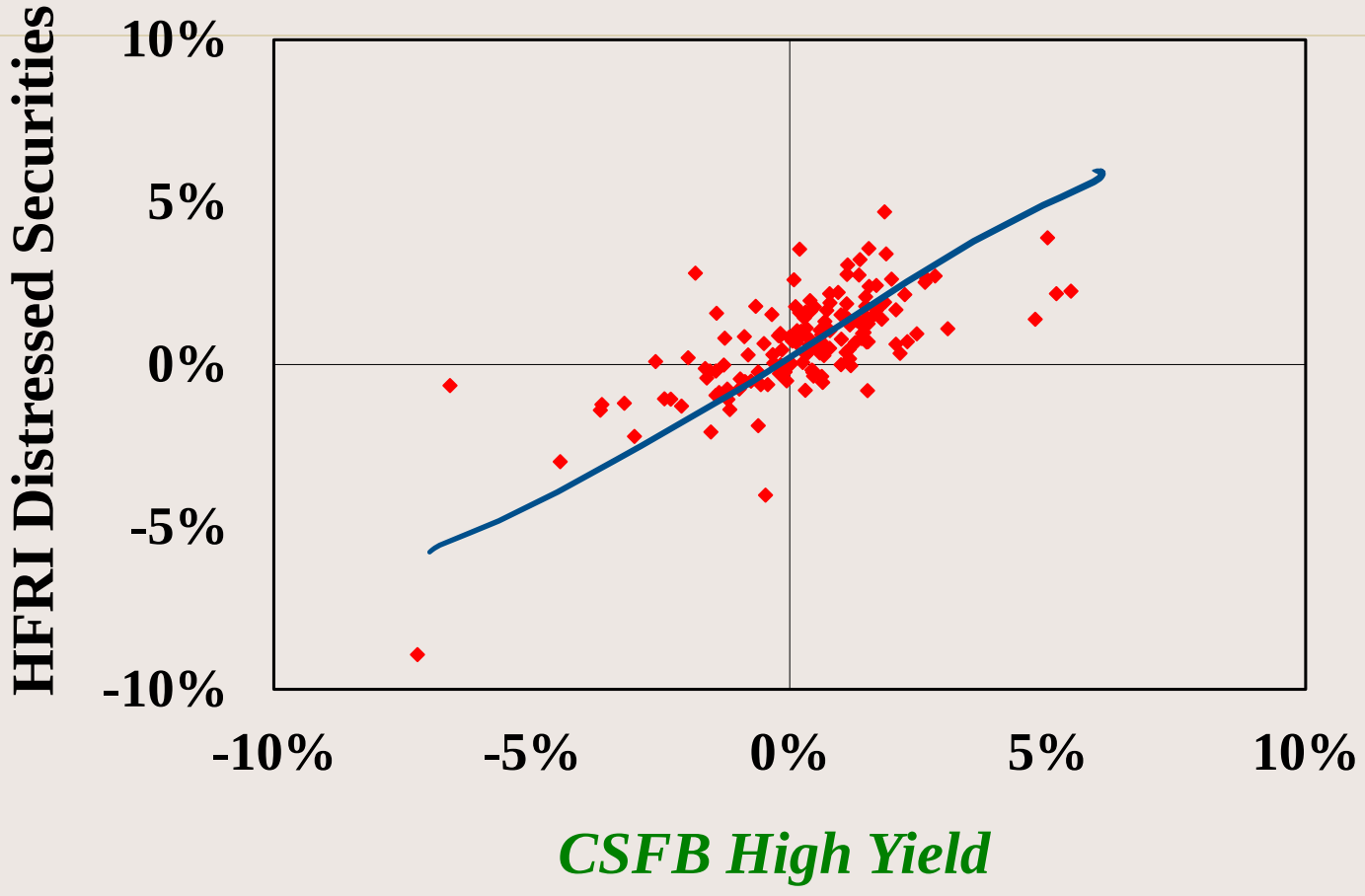
Convertible Arbitrage



Rule-Based Hedged Convertible Bond Portfolio
Long CBs delta Hedged Equity Component

HMC
Exh 5
Long Selection
Zero Coupon Conv
Short Selection
equity

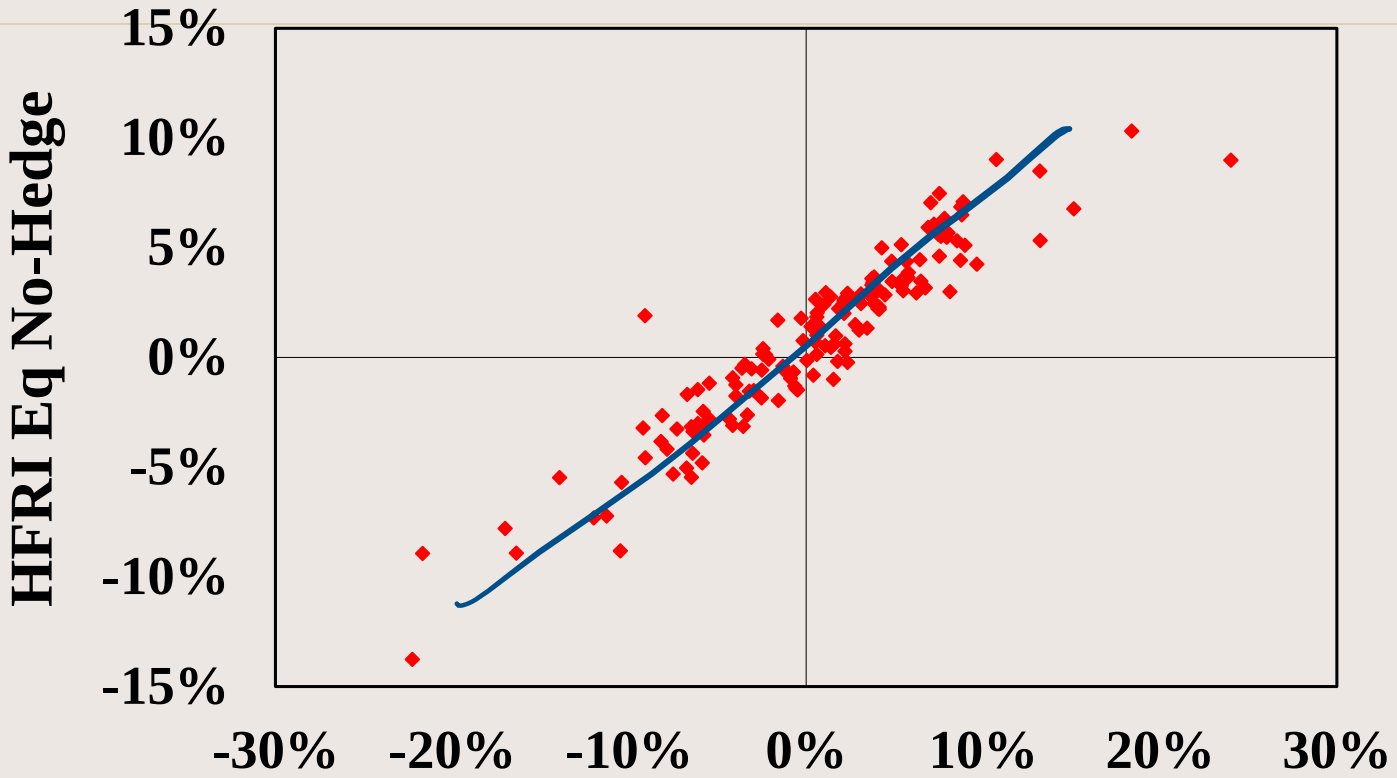
Distressed Investing



Source: Morningstar, HFR

Equity Long-Biased Funds (e.g., 130:30)

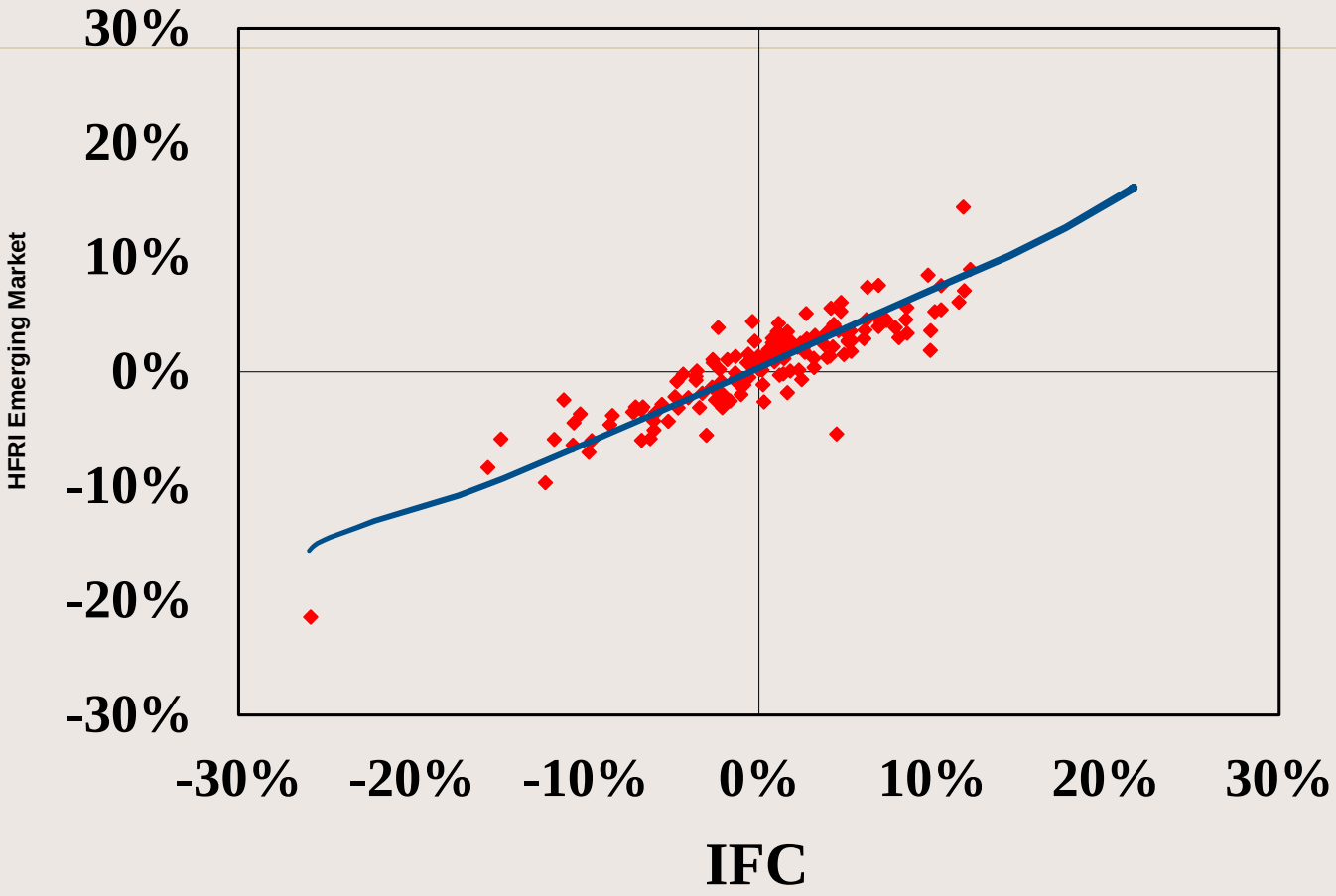
72% — 90%



Wilshire Small Cap Growth

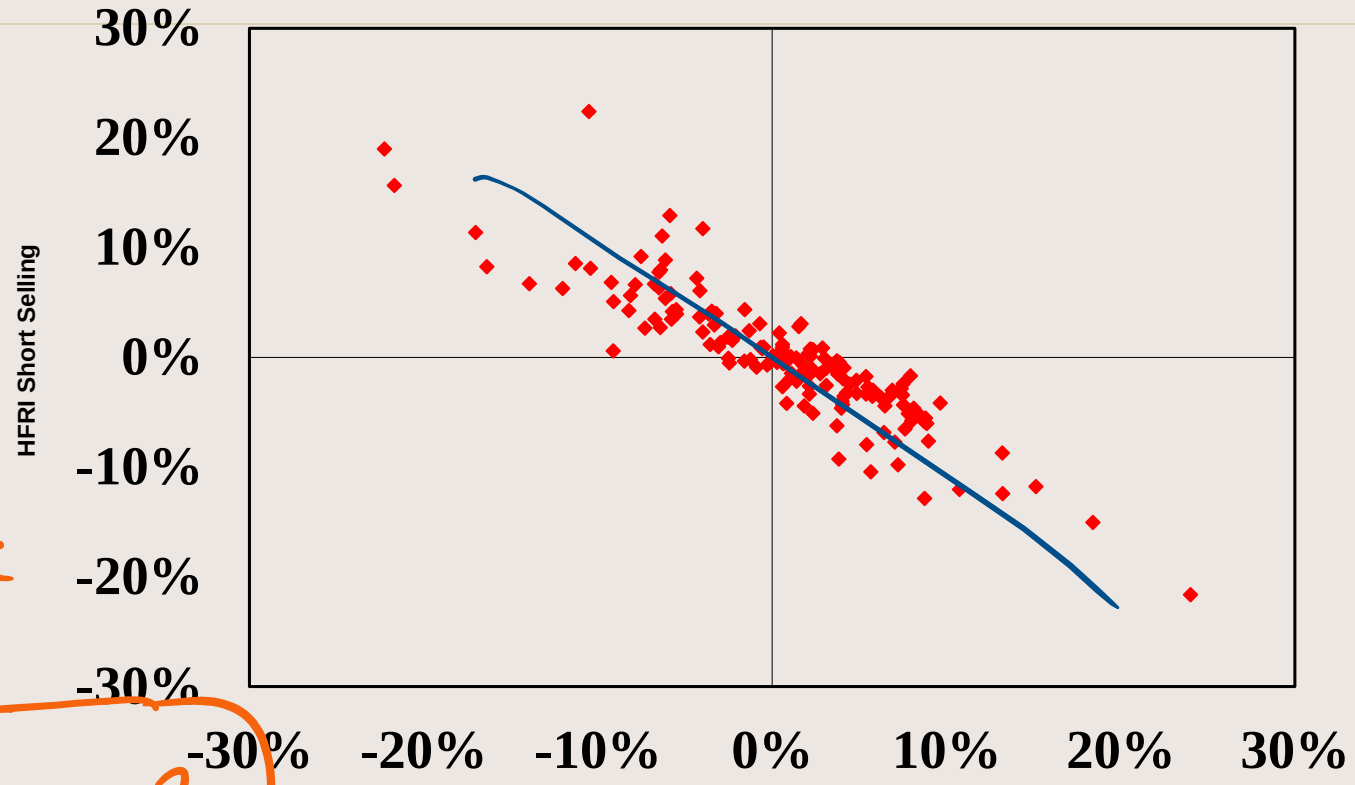
Source: Wilshire, HFR

Emerging Market Hedge Funds



Source: Datastream, HFR

Short Sellers or Dedicated Short-Biased



Wilshire Small Cap Growth

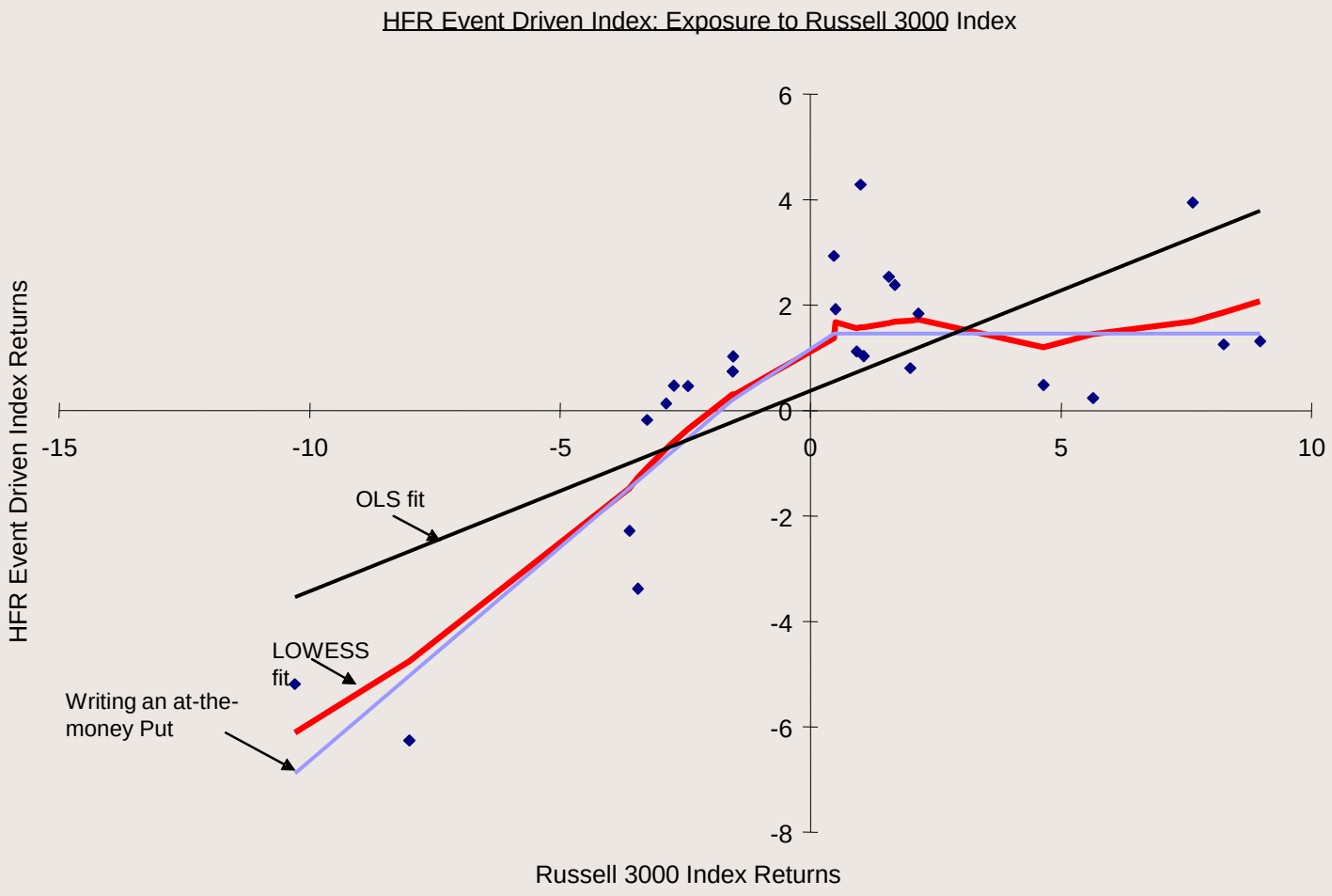
Source: Wilshire, HFR

α

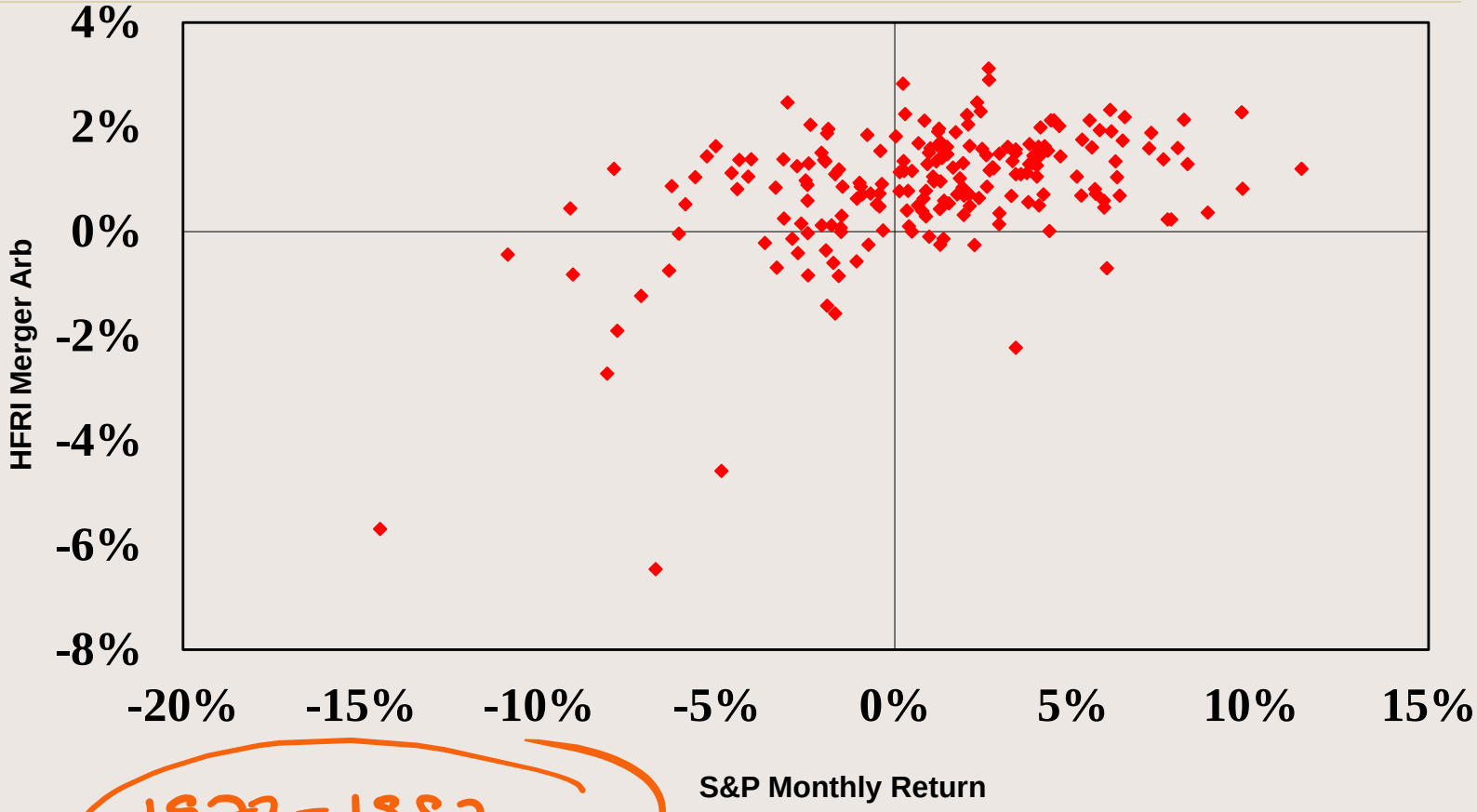
2&20?

Prove for α

Non-Linear Risk in Event Driven HF Strategy



Merger Arbitrage = *Short put option* + *SMB* + *Long Deal Risk*



1973-1992

Source: Datastream, HFR (1990-2006)

Hedge Fund Returns - Mix of alpha and beta bets

$$r = \alpha + \sum \beta_i F_i + e$$

Alternative alpha, Alternative betas

What we would like to have

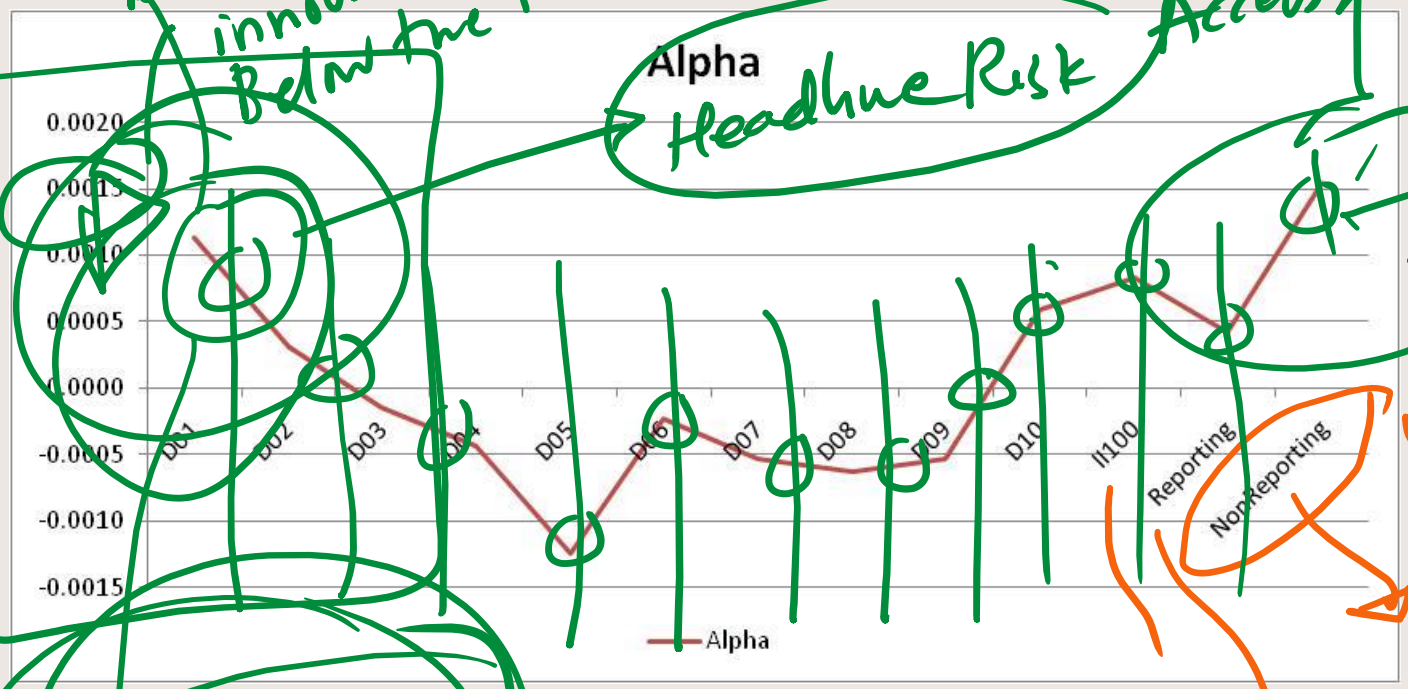
What we need to control for

- F's are hedge fund risk factors: S&P, Small Cap-Large Cap;
- 10-Yr Yield, Credit Spread, Managed Futures Factors, etc

straddles...

out-of-the money put opts.

Alpha Statistics (Deciles 1 to II100 firms)



Incubator
Emerging Managers
New Strategies
innovative
Below the Radar
Headline Risk
Quant models
Access to deals
UBS
Blow up Risk
JFE
Bill Fung + David Hsieh.
CAREER CONCERN

2007

Hello Dolly!

GS ART

- HF Replicators or Clones that deliver alternative beta-based returns and do not carry the high cost of the underlying active management (hedge fund manager fees)

- HF Replicators or Clones that deliver alternative beta-based returns that are rule-based, cheap, with full transparency, high degree of liquidity, institutional quality and free of head-line risk (no Amaranth, no Madoff,…?).

Daily-weekly

60-100 bps

GS
CS
MS
ML

Hello Dolly!

- **Note**: Not all "clones" are born equal, one needs to be careful.
- **Sceptics' View**: Auto pilot can't fly a plane better than a pilot, or can it?

Sources of Investment Returns (documented on a strategy by strategy level)

Top-Down

Un-identified Risk factors

$$r = \alpha + \sum_i \beta_i^{trad} F_i^{trad} + \sum_k \beta_k^{alt} F_k^{alt} + \phi(F_i, F_k) + e$$

Return over LIBOR

= Alpha or skill-based returns

+ Traditional premia and beta returns:

Where you don't want to pay too much fees for....

I-shares

SP500

10YR Treas

+ Alternative premia and alternative beta returns:

Where you have to work a little harder for....

SMB

Δ(BAA-Treasury)

+ Non-linear returns:

options.

Where it is hard to replicate but still doable

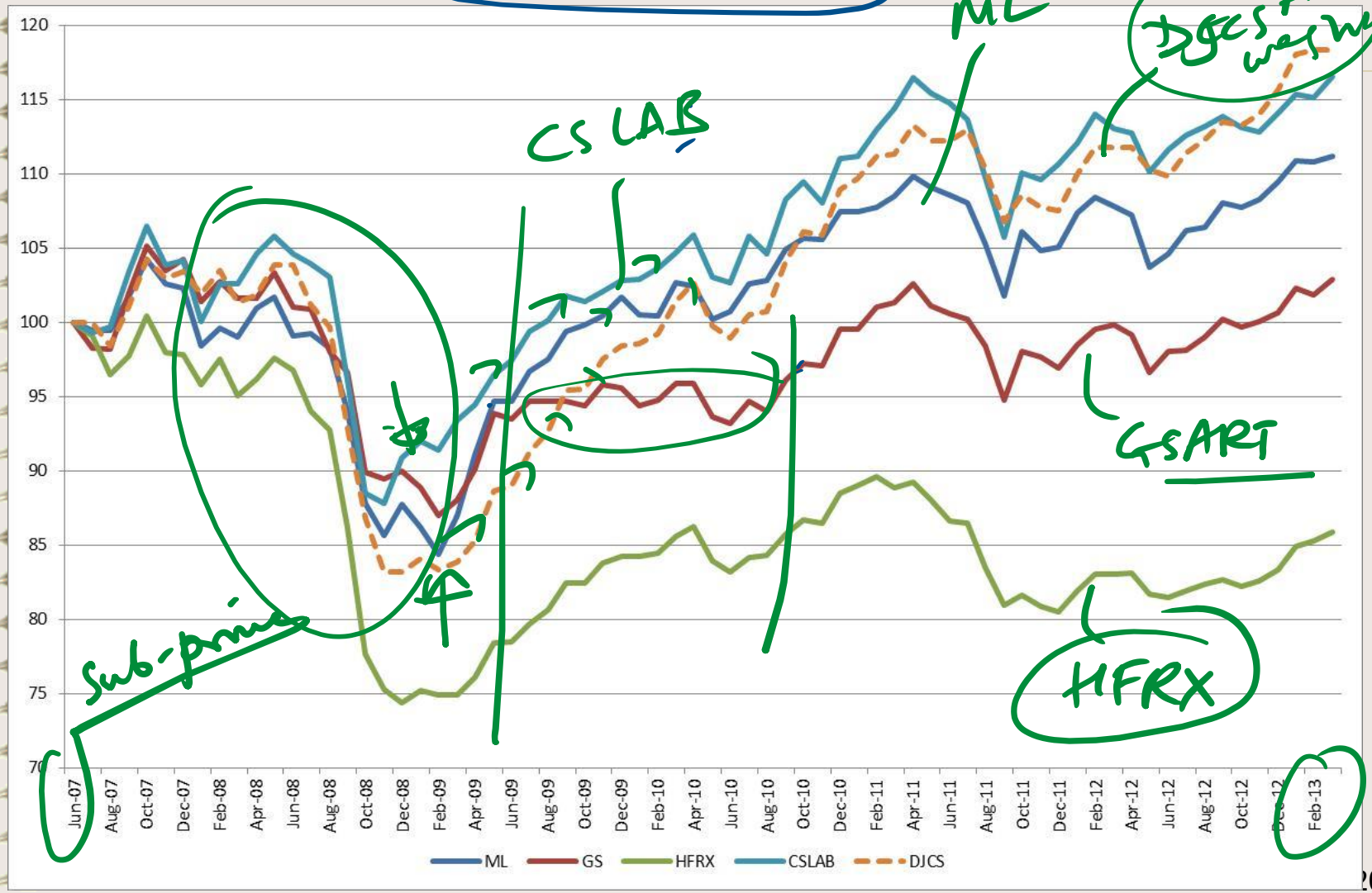
option-based

+ Random returns

Lucky monkey or Un-lucky Einstein

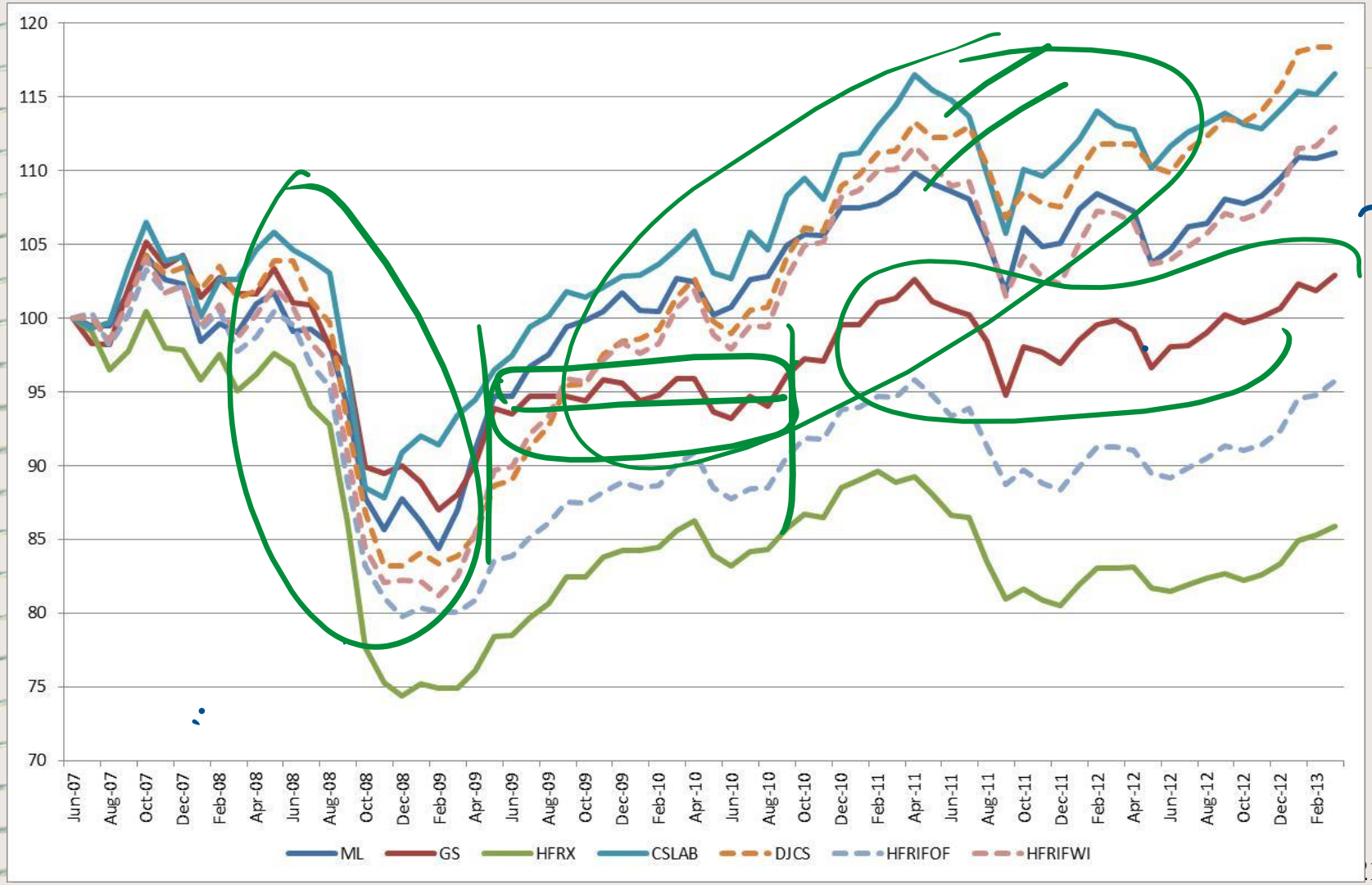
Surely a Robot can't do the job better than the Man, Can it?
Surely an Auto-Pilot can't fly the plane better than the Pilot, Can it?

July 2007 to Feb 2013



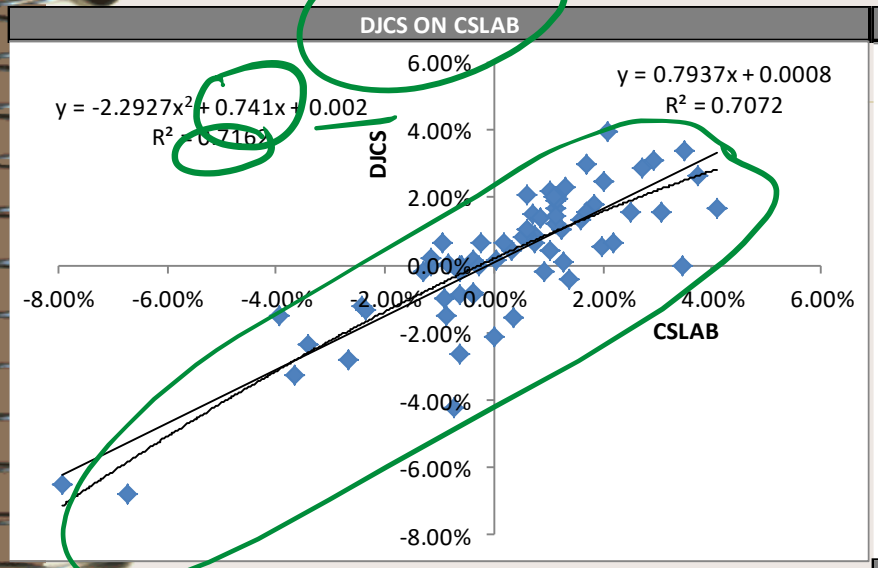
CMINST-1500

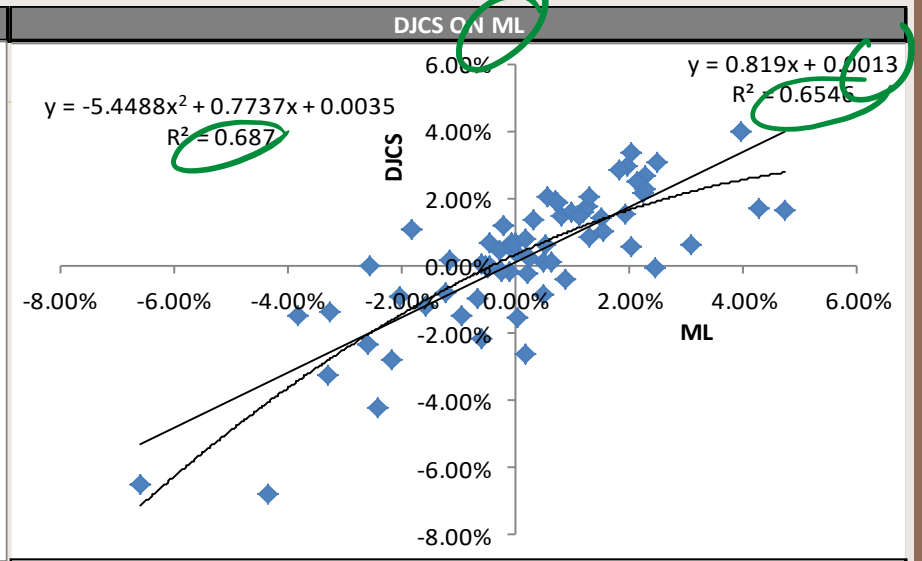
Summary: Performance of HF indexes and the Replicators over recent past spanning the good times and the bad times

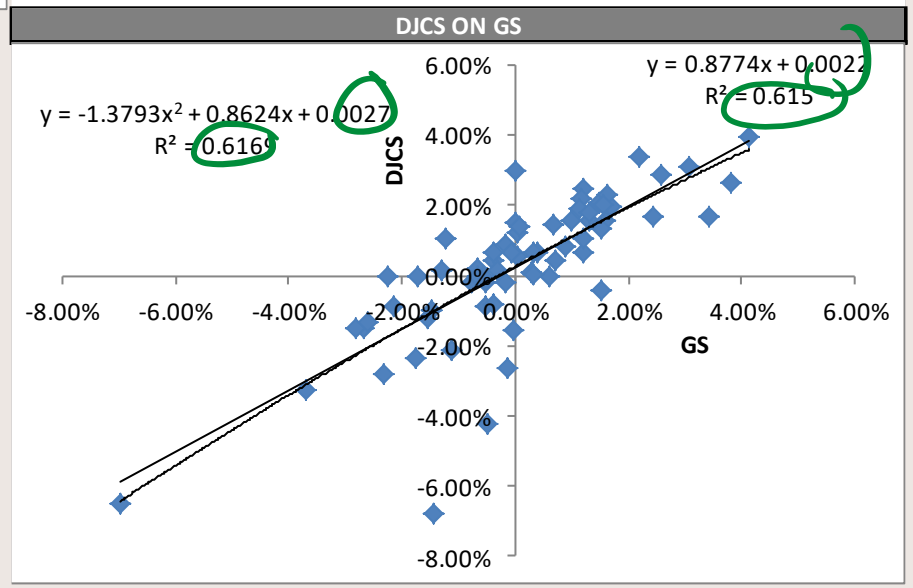


CMINST-1500

DJCS Index vis-à-vis the Replicators







July07-Jan13	DJCS	CSLAB	ML	GS
AVERAGE	0.27%	0.24%	0.17%	0.05%
MEDIAN	0.57%	0.60%	0.23%	0.00%
MAX	3.98%	4.08%	4.71%	4.15%
MIN	-6.78%	-7.94%	-6.61%	-6.98%
STDEV	2.05%	2.17%	2.02%	1.83%
AVG/STDEV	0.13	0.11	0.09	0.03

Source: Bloomberg

Combines real plus simulated performance

With Alternative Beta Replicators, Investors can….

- Identify genuine Alpha producers

α fees

- Direct capital flow towards genuine alpha producers and good alternative beta replicators (Core-Satellite Strategy).

β fees

- "Park the cash" in good replicators till they find alpha-delivering managers (some funds of hedge funds are doing this already).

daily-weekly

- Use replicators to do quick Strategy re-allocation (Long-Short, Event Driven, Macro, etc).

- Use replicators for Risk Management purposes

A New Era has Dawned!

- Until now, investors did not have a credible alternative to accessing Hedge Fund Beta returns in a safe, cheap and passive manner.
- The arrival of Hedge Fund Replication strategies has changed that picture.

liquid Alts

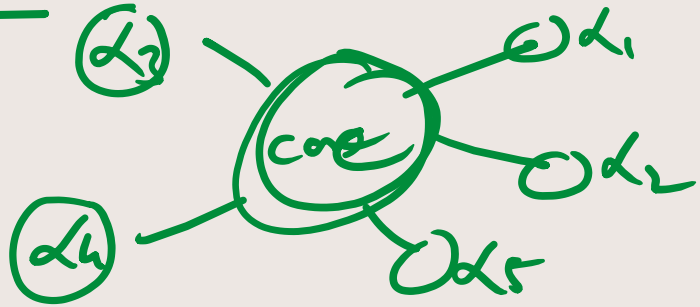
- Now investors can do "Core-Satellite" or "Active-Passive" management in the Alternative Space

HF

- Replicators allow investors to
  - pay "Alternative-Beta Price" for HF Replicators
  - Identify alpha-producing Hedge Fund and Managed Futures Managers
  - Judiciously combine alpha producers with the clones to add value to a well-balanced portfolio

Concluding Remarks

- Unlike mutual funds, hedge funds is a "Tale of Two Risks".
- "Left-tail risk" of Hedge Funds can be offset by combining them with Managed Futures!
- Judiciously selected investments in Hedge Funds and Managed Futures, can add value to traditional Equities, Bonds portfolio and help investors access the risk-premium space in an efficient, risk controlled and cost effective manner.

cost liquid

- By including alpha-producing managers, investors can capture skill-based returns.